9 The stem and leaf diagram shows the ages, in years, of 25 people.

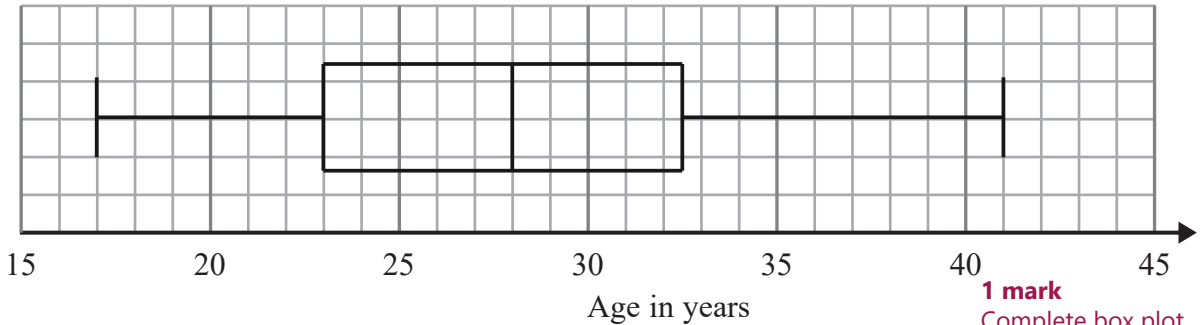
1	7	7	8	9							
2	1	2	4	4	5	5	6	7	8	9	9
3	0	1	2	2	3	4	5	6			
4	0	1									

Key: 1 | 7 represents 17 years

LQ = (22 + 24) ÷ 2 = 23. UQ = (32 + 33) ÷ 2 = 32.5.

Median:13th value = 28; minimum17; maximum41.

(a) (i) On the grid, draw a box plot for this information.



1 mark
Quartiles23,28,32.5

1 mark
Ends17 and41

1 mark
Complete box plot

(3)

One of these people is chosen at random.

(ii) What is the probability that this person is 30 years of age or older?

8 in their30s + 2 in their40s = 10.

1 mark
Count10 out of25

$\frac{10}{25} = \frac{2}{5}$

1 mark
2/5 or10/25

(2)

The grouped frequency table gives information about the ages of a different group of people.

Age (a years)	Frequency
$0 < a \leqslant 20$	7
$20 < a \leqslant 30$	12
$30 < a \leqslant 40$	5
$40 < a \leqslant 50$	1

Anne drew this cumulative frequency table for this information.

Age (a years)	Cumulative frequency
$0 < a \leqslant 20$	7
$20 < a \leqslant 30$	19
$30 < a \leqslant 40$	24
$40 < a \leqslant 50$	25

The cumulative frequency table is **not** correct.

(b) Write down one thing that is wrong with the table.

19 refers to 0 < a ≤30, not 20 < a ≤30.

1 mark

Use cumulative ages

(1)

(Total for Question 9 is 6 marks)